

The methods used to collect the data on Star A were the radial velocity and transit methods. For Star B, only the radial velocity method was employed.

The radial velocity method relies on detecting the small gravitational pull a planet exerts on the star as it orbits the common center of mass. This gravitational force causes the star to wobble slightly. Astronomers can observe this wobble by studying the star's spectrum and measuring the resulting shifts to red or blue (as it moves toward and away from Earth). They will then plot the velocity changes over a certain period to reveal the period and velocity changes that would result from a planetary orbit.

The transit method measures the amount of starlight that is blocked as a planet passes directly in front of its star (if its orbit is favorably aligned with our line of sight). A dip in the light output of a star that repeats with some regularity is characteristic of a transiting planet and allows astronomers to estimate the size of the planet and determine its orbital period.

Based on the above methods, several physical parameters could be determined:

The data collected on Star A allows us to infer a wide range of its properties because both the transit and radial velocity methods were utilized. The amplitude of the star's radial velocity wobbles indicates that the planet must be rather small. In fact, they amount to just over 4m/s. The planet's physical radius and thus volume is inferred from the amplitude of the light drop during transits (from ~ 1.00 to ~ 0.9998). This information is necessary to calculate the density of the planet. Density will tell us whether the planet is of gas composition or rocky composition. From the graphs, we can estimate orbital periods from both graphs. This allows us to roughly calculate orbital distance.

For Star B, we are not able to tell the planets mass, only the minimum mass. We are also able to determine the period (6.34 days) and hence the orbital distance of the planet, we also know that its physical radius and therefore its physical characteristics (apart from size) are unknown due to not using transit data.

Based on our analysis, the exoplanet orbiting Star A is our best bet for finding an Earth-like world and therefore potential life. From the graphs, the radial velocity orbit for Star A indicates that the exoplanet has a mass below 1.5 of that of Jupiter. More specifically from observation of its mass from radial velocity, along with the minute change in luminosity from the transit we can conclude the exoplanet is small and thus must be terrestrial. In addition, its 160-day period would indicate that it likely has moderate surface temperature due to its position away from the center of the Star but close enough to still heat it effectively. Conversely the orbit for Star B has an extremely fast period (6.34 days). With that fast a period the exoplanet must be extremely close to the star. Also given the extremely high amplitude of its orbit the exoplanet must have an immense mass. The exoplanet would be an unsuitably hot gas giant. For this reason, Star A has the better candidate for finding terrestrial, Earth-like life.